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Introduction to Science, Technology, and Society

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The Dialectical Evolution of Scientific Technique

Human practices have purposes. The purposes of most practices are intensely debated. Political science, for instance, questions the role of the social organizations of governments, especially the purpose or goal that governments should achieve. The practice of science differs from other practices by having a widely agreed upon purpose: truth. However, “how” truth is pursued is the controversial topic among scientists. Where politics benefits from a dialectical approach in understanding the evolving goals of politics, a dialectical approach to scientific methods in the pursuit of the goal of science differs. The theses and antitheses of science do not often result in syntheses, but rather a relegation of the opposing methods to an area of science that can most benefit from that unique perspective. In this sense, the dialectic of scientific methods is composed of co-theses. I seek to reconstruct the views of prominent dialectics present in science today and then elaborate further on how these co-theses advance science towards its goal. First, I will establish understanding of natural truths as the indisputable goal of science. Then, I will iterate over the various dialectics and their theses, describing how these theses have advanced science towards its goal. Finally, I will reemphasize how this form of scientific dialectic differs from political or economic dialectic and why this form is necessary for science.

The goal of science itself must be truth. Science for science’s sake has no motive but understanding natural phenomena. Science can be linked to the values and motives of an individual conducting it, such as a drive for fame or compensation, but the historical

constellation that is scientific progress flows towards an understanding of natural processes. That is not to say that scientific progress is linear. It can be slowed and exacerbated, and older forms or theories can resurface and make connections with contemporary trains of thought. Still, science marches forwards towards its goal. The innovations and technology developed and the theories yet to be disproven support this.

The most controversial arguments within science concern the nature of humans and their role in producing objective knowledge. It is from these human-centered controversies that the co-theses develop. “Saints versus angels” shifts scientific practice from a paradigm of rational epistemology to a slightly psychoanalytic framework of the performance of science by humans. Science, as a human-created and human-practiced activity, must be looked through the lens of human perspective. “Saints versus angels” speaks towards a larger opposition within science of truth-to-nature and mechanical reproducibility. This dialectic poses the expert knowledge of “angels” against the virtuous methodology of “saints.” In “The Image of Objectivity,” the authors use atlas makers as a microcosm to represent this larger scientific issue. Before photographs, atlas makers were sorted into categories: naturalist and idealist. The naturalists sought to recreate the details of nature exactly how it was in front of them. The idealists used their judgment to create a golden mean of what they believed to represent every instance of an object most closely. The divide in this argument widened with the advent of photography. The naturalists turned towards photography to ‘objectively’ capture nature. Ironically, both sides imposed their own subconscious ideals onto their subjects; the naturalists with which subject they chose for their photo, and the idealists with the way they balanced what they deemed imperfections. The reason these are co-theses, and one practice did not wipe out the other, is because they were both given their respective places within a scientific framework. Mechanical

reproducibility was relegated to hard sciences such as physics, and the expertise of idealists was adopted by soft-scientists who found difficulty employing experiments the same way hard science did – namely psychology. The multifaceted nature of scientific truth allowed for both of these theses to remain within practice.

Another dialectic occurred within how scientific studies conducted their method of research, namely the practice of fastidious versus pragmatic trials. As discussed in "The Construction of Lay Expertise: AIDS Activism and the Forging of Credibility in the Reform of Clinical Trials," fastidious trials aim to minimize variables in order to get purer data points that can provide information concerning the true nature of phenomena. Pragmatic trials, on the other hand, took an approach towards phenomenological truth as existing within a complex system of variables. In the realm of science, this approach could be said to emphasize the application of knowledge in order to glean more knowledge. The dialectic was one of normative science versus positive science; theory versus reality. For AIDS research, pragmatic options were favored by realizing that the most accurate experiments could be achieved by human subjects who were not pure or clean. The law of large numbers certainly played a role in coming to this conclusion as 'imperfect' subjects could be more easily acquired en masse. That is not to say that fastidious trials have no place in science; rather, that method of inquiry was better suited to long-term studies with a goal of theory than short-term studies aiming to be applied as soon as possible to achieving solutions. Both approaches contribute to understanding, but the aforementioned value assessments deemed those approaches most effective in different areas of study.

Engaging with science as a human based activity also raises the question of discovery as either an objectively mechanistic view of the world or a constructed view based on human perspectives. In "A New Window on the Universe: The Non-Detection of Gravitational

Radiation," the author describes the dialectic of scientific realism against constructivism. Realists imagine human theories to exist objectively, blending themselves to reproducibility within an objective reality. On the other hand, constructivists see the social construction of knowledge as the critical practice towards truth. Within the paper, peer reviewers hold this constructivist point of view while also appealing to a realist ideal of replicability. This dialectic differs from the previously mentioned ones by the applicability of both theses within singular vestiges of epistemology.

Finally, the dialectic of flexibility versus rigor within scientific methods emphasizes the mosaic of possibilities for forms that science can take. In "A long journey to reproducible results," the authors contrast the benefits of consistent reproducible methods with the ability to pivot around real-world complexities omnipresent in any study. Protocols and well-tested methods may not be possible to execute perfectly within every realm of science, especially one with as many inherent variables as biology and medicine. The idea of a perfectly deterministic scientific study is enchanting, but it is impossible for any researcher today to adhere and execute on this fairytale. In that sense, to strive for this possibility as a target to aim for has been adopted, and the ability to be flexible in one's methods of discovery is equally beneficial within a realist perspective.

Dialectics impact science in a different way than those of economics or politics. This difference is due to how the scientific goal of accumulating knowledge will remain unchanged due to exactly how science is defined. Every epistemological theory has a place in epistemology because every bit of new information is additional information, and every method of acquiring information acquires information. The value in place remains unchanged.